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Multiagent simulations of hunting wild meat in a village in eastern Cameroon

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Abstract

This paper is based on a study of blue duiker hunting in Djemiong, a forest village in eastern Cameroon. The aim of the study is to understand how the organization of the hunting activity between villagers constitutes a management system. The main species hunted is a small antelope, the blue duiker (*Cephalophus monticola*, Thunberg). Surveys were conducted to understand the inhabitants' hunting behaviour. Hunting takes place 6 months/year and is subject to a spatial shifting rule. Every year, each hunter changes the location of his traps. This behaviour is presented as a management rule that is implemented by the hunters. A spatially explicit individual-based model is used to compare different rules for trap locations in space and time. We propose a model based on the use of multiagent systems. Creating a multiagent system involves reproducing an artificial world that resembles the observed world — i.e. it is made up of different actors — for experimental purposes. Each agent is represented as an independent computerized entity capable of acting locally in response to stimuli or to communication with other agents. CORMAS (Common-pool resources and multiagent systems), a generic simulation environment based on SMALLTALK, makes it possible to build flexible spatially explicit individual-based models. Using this multiagent simulation software, a model was built based on the life history of the blue-duiker and on the inhabitants' hunting behaviour. The model incorporates data from a geographical information system (GIS) to create an artificial landscape that resembles the village landscape. The results highlight the importance of coordination between hunters, particularly the fact that trap networks are appropriated by family groups. The spatial location of traps seems to have a much more crucial influence on the model than global hunting pressure and the duration of the close season. © 2001 Elsevier Science B.V. All rights reserved.

Keywords: Duiker (*Cephalophus monticola*); Hunting; Multiagent system; Population dynamics; Spatially explicit individual-based model; Wild fauna management

1. Introduction

The wild fauna in Africa is one of the richest and most varied in the world. Wild fauna has been used everywhere from time immemorial, yet it is rarely considered to be a consumable re-

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source. Its importance is often reduced to its great tourist appeal (Chardonnet, 1995). However, hunting wild animals provides meat for the family and an additional source of income for people in sub-Saharan Africa. This applies to both 'hunter-gatherers' and so-called 'agriculturalists'. The latter group only use wild resources to complement their crops and domestic production, particularly during the hungry gap that usually occurs between two crops (Chardonnet et al., 1995). Traditional hunting is, therefore, essential to farmers in a number of African countries albeit on a small scale (Bissonette and Krausman, 1995).

This paper is based on a study carried out in Djemiong, a forest village in eastern Cameroon. There are no protected areas in the region and the challenge is to understand how the resource is managed. The aim of the research is to develop a model for studying the viability of the management system. The main species hunted is the blue duiker (*Cephalophus monticola*, Thunberg). Surveys were conducted so that the inhabitants' hunting behaviour could be understood (Takforyan, personal communication).

Data on the harvest rates of species hunted for meat as well as on the maximum possible production per unit area of forest are needed to estimate the impact of hunting on wildlife populations. Mathematical models are traditionally used for issues relating to wildlife resource management (Robinson and Redford, 1994). The aim of this study is to compare ecological production with an estimated harvest. Therefore, the model used should include estimates of reproductive productivity and population density. The conclusions drawn from previous studies are often alarming (Robinson and Bodmer, 1999). However, in tropical forest conditions, estimating existing hunting yields and maximum sustainable harvest rates is no easy. In Cameroon, for example, reports suggest that blue duiker harvest rates are as much as 25 times higher than estimated sustainable levels, yet hunting continues more or less unabated matter (Inamdar et al., 1999).

Beissinger and Westphal (1998) have recently reviewed the structure, data requirements, and outputs of a wide range of models used in population viability analysis: analytical, deterministic single-population, stochastic single population, metapopulation and spatially explicit models. The choice of the most appropriate model depends on the object of the study.

In Djemiong, the resource is hunted 6 months/year and there is a spatial shifting rule. Every year, each hunter moves his traps to a different location. This practice is presented as a management rule implemented by the hunters. The key question is: can the rule that regulates access to space at different times of year be considered as a management rule? To answer this, we need to carry out a spatial and dynamic simulation of the resource component that takes into account the hunters' behaviour and the way they interact when they decide to locate their traps on village land.

A spatially explicit individual-based model seems suitable for the spatial and dynamic simulation of the resource component. We use multi-agent systems (MAS) to deal with the problem of coordination between the hunters. MAS — computer formalism from the field of distributed artificial intelligence — are generally used to study interactions between the basic components of a system. MAS are very similar to spatially explicit individual-based models when it comes to building simulation models that allow the movement, mortality, and reproduction of each computer entity to be monitored on an artificial landscape (spatial grid).

2. Methods

We start by introducing the principles of MAS and then go on to describe our multiagent simulation software CORMAS (Common-pool Resource and Multiagent Systems, Bousquet et al., 1998). We use this generic tool to develop a model based on the blue duiker's life history and the inhabitants' hunting behaviour. The individual-based model of the duiker integrates data from a geographical information system (GIS) so that the village landscape can be simulated artificially.

2.1. Multiagent systems

The modelling approach that we propose is based on the use of multiagent systems (Ferber, 1999). In order to model complex phenomena, multiagent systems represent agents of the observed world and their behaviour. Creating a multiagent system involves reproducing an artificial world that resembles the observed world, in that it is made up of different actors, for experimental purposes. Each agent is represented as an independent computerized entity capable of acting locally in response to stimuli or to communication with other agents. When simulating populations of agents, emphasis is generally placed on the notion of the environment. We examine the relations between agents via their actions on the environment. Today, ecologists and social scientists are developing multiagent simulation methods using a wide variety of applications and several methodological studies (Villa, 1992; Congleton and Pearce, 1997; Weiss, 1999). Multiagent systems provide a methodological approach and are effective tools for implementing spatially explicit individual-based models. Swarm, an agent-based simulation toolkit created at the Sante Fe Institute (Minar et al., 1996) is mentioned by Lorek and Sonnenschein (1999) in their survey on software programmes for implementing individual-based models.

2.2. The CORMAS software

Our research team is studying renewable resource management, which depends on understanding the interactions between natural and social dynamics. We use a simulation methodology involving MAS in order to understand the complexity of these interactions. Models of this type have been developed for irrigated land management in Senegal (Barreteau and Bousquet, 2000) and herd mobility in the Sahel (Bousquet et al., 1999a). In association with these research projects, tools have been developed and an approach to modelling has been proposed (Bousquet et al., 1999b). We have developed a simulation environment called CORMAS (Common-pool Resources and Multiagent Systems; Bousquet et al., 1998) that combines these tools.

CORMAS has been developed to provide a multiagent framework that can be applied to the interactions between a group of agents and a shared environment. It aims to simplify the task of simulating resource management. CORMAS was developed using the software VISUALWORKS. It uses and proposes SMALLTALK as a development language. Using object-oriented programming, CORMAS provides pre-defined generic entities from which the specific agents inherit (this applies to any model built using CORMAS). CORMAS is freely distributed (http://www.cirad.fr/presentation/programmes/espace/cormas) and comes with a library of models (including the blue duiker model).

2.3. The blue duiker model

2.3.1. The artificial landscape

We have developed an artificial landscape similar to that of Djemiong village. The limits of the village were defined on a map in consultation with the inhabitants. Some data has been digitalized with a GIS and set in raster format. The spatial resolution is 3 ha, given that this is the average range of the blue duiker. The GIS is divided into three layers: roads, rivers and hunting localities. A file is created for each layer to provide information on each cell. CORMAS then imports this data. The cell has an attribute for each layer: water (yes/no), road (yes/no) (cf. Fig. 1a) and the hunting localities (cf. Fig. 1b).

2.3.2. The model for population dynamics

Different data was collected to simulate the life history of the blue duiker. Dubost (1980, 1983a,b) is the main source of information. Blue duikers are territorial animals. Once they have reached maturity and found a mate, they demarcate their territory where they remain until they die. The average size of the territory is 3 ha, which is the spatial resolution for the model, i.e. each cell on the spatial grid (Fig. 1a) represents an area of 3 ha.

A duiker agent was created that takes into account age, sex, gestation length and mate. The blue duiker agent's behaviour — growth, mortality, and reproductive functions — is determined

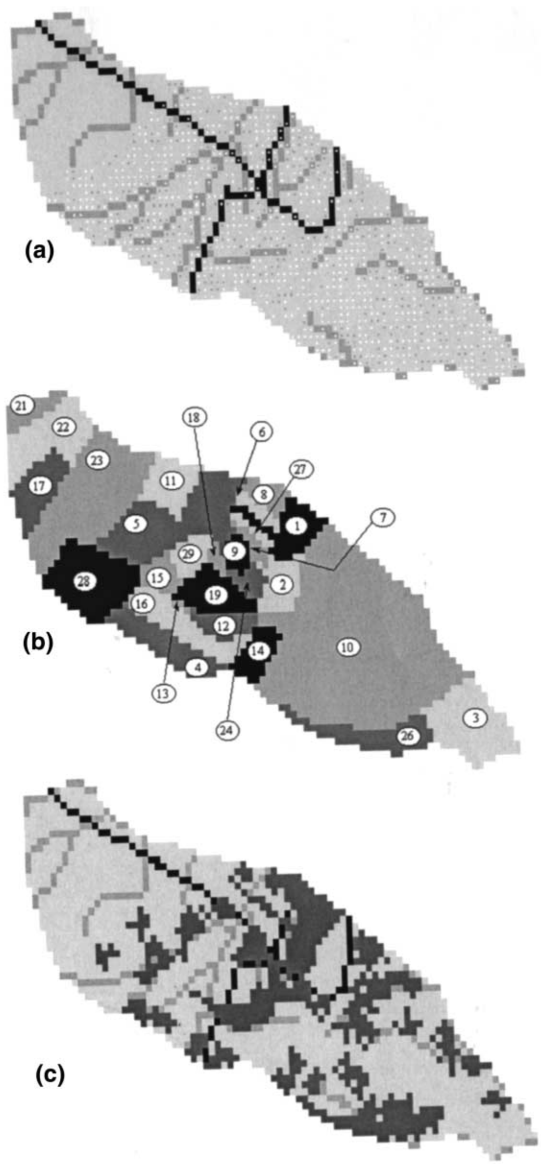

Fig. 1. (a) The Djemiong artificial landscape. Each light grey cell represents an area of 3 ha, which is the size of the adult duiker's territory. Cells with water are dark grey, cells with a road are black. The small dark grey dots represent adult duikers with a mate and the small white dots represent juveniles or single adults. (b) The 27 hunting localities in Djemiong are each identified by a number. Table 2 summarizes some of their characteristics. In the upper part of the map, localities 17, 21, 22 and 23 were not surveyed and they are, therefore, not included in the model. (c) Location of the trap networks on the model's spatial grid for experiments R1 and R2. The first layer (light grey) represents standard cells. The second layer represents water (medium grey), the third represents a road (black) and the last one (dark grey) represents the traps.

weekly (time step = 1 week) by the above parameters and by a number of rules that govern movement (cf. Table 1: model parameters).

2.3.2.1. Growth. The model's time step is 1 week, which means that all the temporal attributes or parameters are expressed in weeks. Thus, the age of each duiker agent is simply increased by 1 at each time step. On this basis, weekly age classes could be defined by grouping together duiker agents of the same age. However, in this example it is sufficient to classify age in years (not weeks) to describe the population's age structure (cf. Figs. 7 and 8).

2.3.2.2. Mortality. The natural mortality of young blue duikers (0–40 weeks' old) is 30%. This group's survival rate is converted into a probability for weekly natural mortality (M_y), where:

$$M_y = 1 - 0.7^{1/40}$$

In the same way, two other parameters were set for the natural mortality of adults ($40 \leq \text{age} < 240$) and old individuals ($\text{age} \geq 240$). The blue duiker has a life expectancy of 6 years, which corresponds to the threshold of 240 weeks. The survival rate of old individuals (240–340 weeks) is only 1% and, therefore, the corresponding probability for weekly natural mortality (M_o) is equal to:

$$M_o = 1 - 0.01^{1/100}$$

We assume that the survival rate of adults (40–240 weeks) is high (95%). This is expressed as:

$$M_a = 1 - 0.95^{1/200}$$

2.3.2.3. Movement. A juvenile duiker reaches maturity at the age of 72 (female) and 95 weeks (male), at which point it leaves the parental territory to find a mate and establish a territory of its own. Blue duikers are monogamous: a pair of animals stays together in the same place until one of them dies. Thus, in the model, only mature and single duiker agents move.

The perception range of a duiker agent is defined as a three-order recursive function based

Table 1
Model parameters

Parameter	Condition	Value	Reference
Cell area (territory size)		3 ha	Dubost, 1980
Weekly probability for natural mortality			
M_y (young individuals)	age∈[0; 40]	$1–0.7^{1/40}$	Dubost, 1980
M_a (adult individuals)	age∈[41; 240]	$1–0.95^{1/200}$	Chardonnet, 1995
M_o (old individuals)	age > 240	$1–0.01^{1/100}$	
Weekly probability for hunting mortality	Area with traps	0.15	Ngandjui, 1997
Age at maturity	Male	95 weeks	Chardonnet, 1995
	Female	72 weeks	
Gestation length	Female	30 weeks	Chardonnet, 1995
Parental care period		21 weeks	–
Hunted area with trap network		12 or 36 ha	Ngandjui, 1997
		i.e. 4 or 12 cells	Takforyan, personal communication

on the four-connex neighbourhood of the cell where it is located. In a weekly time step, a duiker agent can visit any of the 25 cells that make up this area. When a single adult male meets a single mature female, they look for a suitable cell within their common perception range where they can settle. The suitability is defined as follows: the new site should be empty (no other duiker agents present) and have neither water nor road.

When mature and single duiker agents fail to find a mate and a suitable territory, they choose a site randomly from the empty cells (that do not already have a duiker agent) in their perception range. These rules are summarized in Fig. 2 and the resulting spatial distributions for the duiker agents are similar to those shown in Fig. 1a.

2.3.2.4. Reproduction. A female duiker agent becomes gravid as soon as she has found a mate and established a territory. After a 30-week gestation period, a newborn duiker agent is created. The newborn duiker stays in its parents' territory until it reaches maturity. The female duiker agent cannot conceive again until its young is 21 weeks' old. This parameter is defined in the model as a parental care period. The model is calibrated so that the average interval between two successive births is about 13 months (the value found in references on the subject).

The preliminary experiment in the next session involves the simulation of this biological and ethological model.

2.3.3. The hunting model

A survey was conducted in the village to find out where hunters have been hunting for the last 11 years. Twenty nine hunting localities were identified from the spatial information collected in the survey (cf. Fig. 1b and Table 2).

In Djemiong, hunting only takes place during the gap in production that occurs between two farming seasons. In the model, the farming season lasts for 26 weeks each year, during which time traps are removed. This is followed by a 26-week hunting season. During the hunting season, each hunter sets traps along a path in the forest (trap network). The network covers an area that ranges from 25 to 100 ha. If we assume that the range is 25 ha in the model, a trap network extends across eight cells (3 × 8 ha). In fact, the results from the field survey suggest that there are two kinds of hunting localities. The hunting localities 1, 2, 3, 5, 10, 11, 12, 14, 26 and 28 (cf. Fig. 1b) are the largest and furthest from the village (cf. Table 2). These areas are mainly used for hunting. In general, they are more heavily hunted than the areas closer to the village (located at the crossroads, near locality 9, cf. Fig. 1a and b). These areas are used for farming for 6 months of the year. This heterogeneity is represented by the distribution of trap networks, which extend across 12 cells in the larger hunting localities and four cells in the smaller ones (cf. Table 2).

The probability of catching a duiker in an area that is hunted is 0.15. This is used as an additional probability for weekly mortality, H. We know little about the relationship between age and the likelihood of being caught, so we assume that H is independent of age.

2.3.4. Overall model for Djemiong

A general flow chart (Fig. 3) shows how the hunting activity is added to the biological model. The grey boxes represent the steps that are affected by the addition.

The hunting model involves setting and removing traps. The initial duiker agents' population, i.e. the population to be hunted, is established from a structured and stabilized population obtained by running the biological model for 100 years. The hunting activity obviously effects the mortality function and this is expressed by an additional mortality probability.

3. Experiments

The aim of the model is to determine how the location of trap networks effects population viability and the number of animals caught in each hunting season. The experimental plan outlined below has been drawn up for the simulation.

3.1. Preliminary experiment: model of blue duiker population dynamics

We simulate the evolution of the system with no hunting for 5000 weeks (nearly 100 years) in order to calibrate the model of population dynamics. Ten different values for initial population

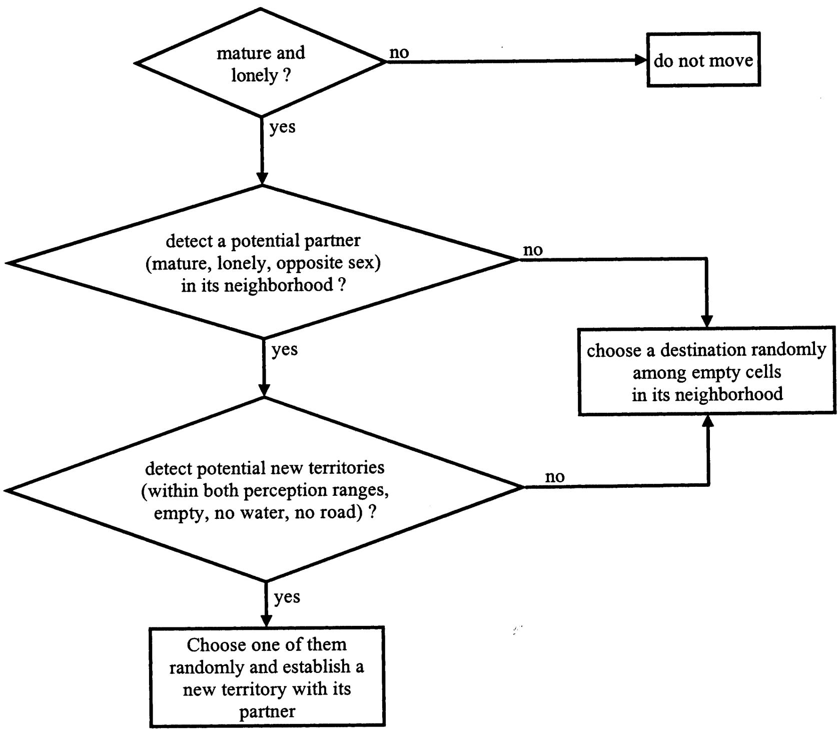

Fig. 2. Flow chart for the duikers' movements. The migratory behaviour of the duiker agent is such that "if conditions allow, then statement 1 applies, otherwise statement 2 applies".

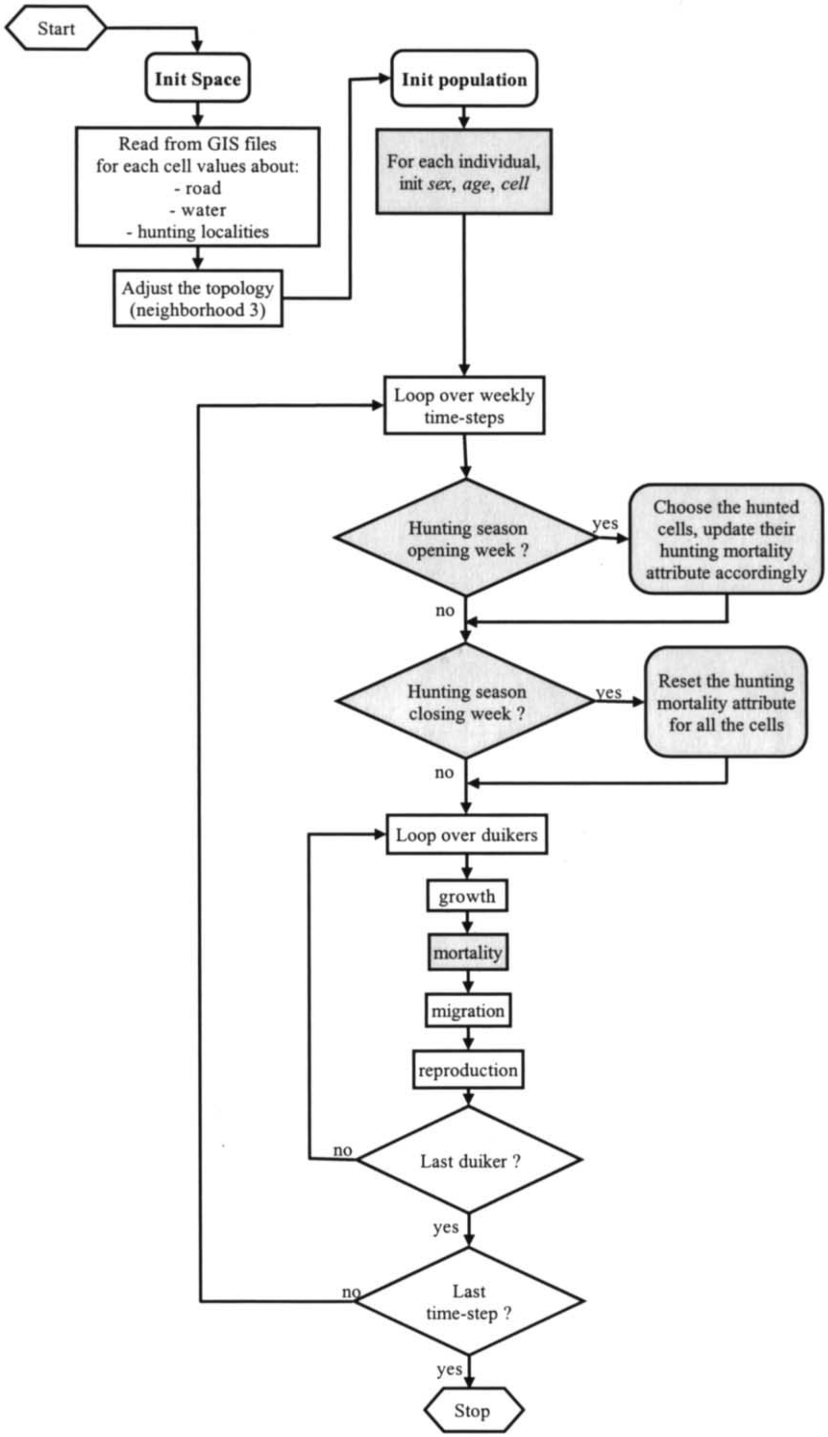

Fig. 3. Overall flow chart for the Djemiong model.

Table 2
The hunting localities in Djemiong village

Hunting locality	Total number of cells	Total number of suitable cells (no water, no road)	Maximum size of trap network (number of cells)
1	59	48	12
2	44	31	12
3	117	106	12
4	71	59	4
5	159	142	12
6	15	12	4
7	14	10	4
8	25	16	4
9	19	11	4
10	596	522	12
11	55	44	12
12	33	24	12
13	3	3	4
14	39	35	12
15	26	18	4
16	71	54	4
18	9	5	4
19	58	45	4
24	19	10	4
26	71	60	12
27	10	7	4
28	116	96	12
29	29	24	4
Total	1658	1382	

abundance were randomly chosen and then tested. Depending on the value for population abundance, duiker agents were created by attributing their sex with a ratio 1, their age at random (from 0 to 80 weeks) and their location at random within the spatial grid.

From this preliminary experiment the population of duiker agents is recorded after 5000 weeks have elapsed. This value is used as the starting point for all the experiments that involve hunting activities.

3.2. First set of experiments: repeating the data from the 1995 hunting season

Data from the field survey reports 90 hunting actions in the corresponding hunting locality for the year 1995. In the model, this value is taken to be the standard hunting pressure. Each hunting action switches a set of four or 12 cells to a 'hunted' state. If the space available in a given hunting locality is insufficient (i.e. total number of cells is less than four or 12 times the number of hunting actions reported), all the cells in the hunting locality are switched to the 'hunted' state.

We begin by using the model to test the effect of the periodicity of the hunting activity. This feature is considered as a classical management tool. The first scenario, R1, involves the continual repetition of data from the 1995 hunting season (cf. Fig. 1c). In the second scenario, R2, the traps are removed from the spatial grid every 26 weeks and put back again in the same place (Fig. 1c) every 6 months. This 6-month periodicity corresponds to what actually happens in Djemiong. All of the subsequent experiments include this feature. In fact, we have no information about the precise location of a trap network in a hunting locality. As in scenario R2, it seems reasonable to assume that the trap is located in the same place each time a hunter decides to access a given locality. This is presumptuous because the information obtained from the survey is not on the same scale as that used for the hunting process in the model, which was determined by ethological processes. To test the effect of this assumption, a third scenario, R3, was outlined. It is the same as the R2 scenario except for the fact that at the beginning of the 6-month hunting season, each trap network is randomly relocated in the same hunting locality.

The above experiments are not very realistic because they suggest that traps are always located in the same place, as in the 1995 hunting season. According to the field survey, hunters actually avoid using the same hunting locality from one season to another as a way of preserving resources. This aspect is taken into account in a second set of experiments in which the model incorporates the hunters' behaviour.

3.3. Second set of experiments: hunter agents determine the location of trap networks

Ninety hunter agents were created. Each one has a collection of four precise trap networks, which is randomly initialized from a set of four hunting localities. The probability of a hunting locality belonging to this collection depends on its relative size (cf. Table 2), i.e. the larger hunting localities are used by more hunter agents than the smaller ones. The first scenario, H1, states that a hunter agent chooses his trap network randomly from the three not used during the previous season.

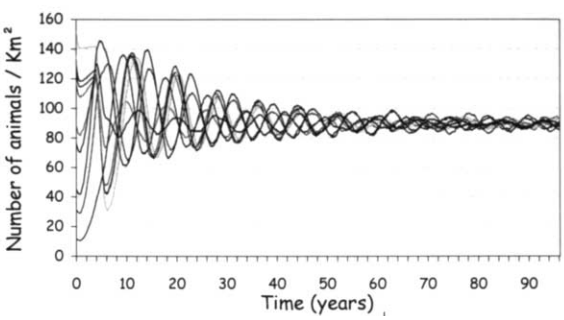

Fig. 4. Evolution of the population density during 5000 weeks with no hunting. For the ten runs, the sex ratio of the initial population is equal to 1, the number of duiker agents is random and their age is attributed randomly between 0 and 80.

In this scenario, hunters act individually (individual turnovers) and there is no coordination between them. In the second experiment, H2, 30 groups of three hunter agents are defined. Four specific trap networks are assigned to these groups or collective entities, which represent kinship groups (i.e. hunters from the same family). The rule for individual turnovers also applies here, but with an additional constraint: a trap network cannot be hunted by any of the three group's members for three successive seasons.

4. Results

All the experiments were run ten times to test the constancy of the results and to calculate the mean values and S.D.

4.1. Preliminary experiment: the model for blue duiker population dynamics

The results for population density (Fig. 4) suggest that the model stabilizes when there are approximately 90 animals/km^2 with slight oscillations (for ten runs, the average population density is 89.53 after 5000 weeks and the S.D. is 2.11).

This is an important result because it is the density observed in the region's of forest where there is no hunting. Thus, by simulating the behaviour and the interactions of the animals at a microscopic scale we observe a population property at a global scale. This is an empirical validation of our individual-based model. The oscillations are due to the fact that the population was initialized randomly, particularly for age, so to start with there is a very large group of young duiker agents. Progressively, the population's age structure reaches a steady-state equilibrium determined by the three age-dependent probabilities for natural mortality (cf. Fig. 7a).

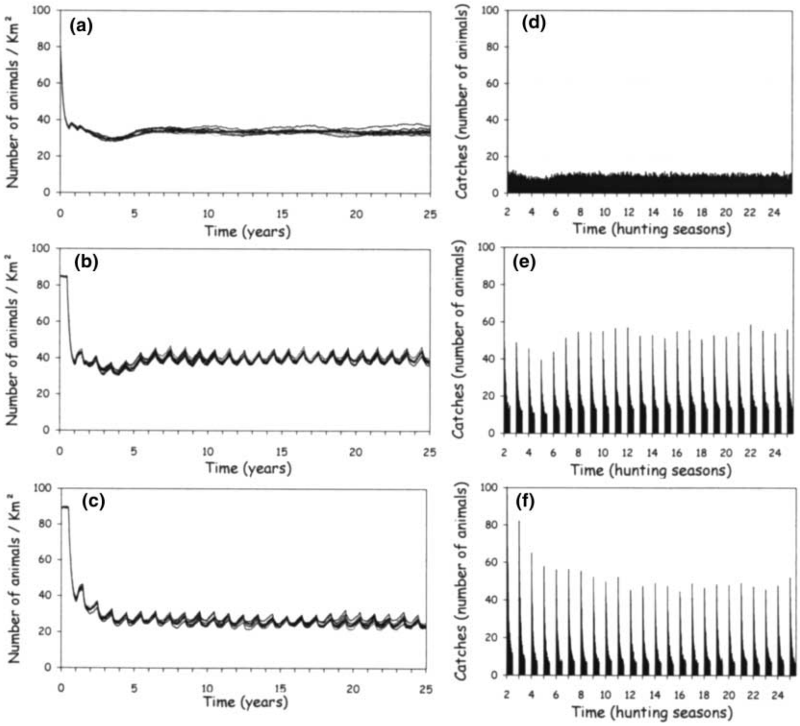

Fig. 5. (a) Evolution of the population density during 1300 weeks of simulation with scenario R1: continuous hunting and fixed trap locations (cf. Fig. 1c). (b) Evolution of the population density during 1300 weeks of simulation with scenario R2: periodic (every 6 months) hunting with the same trap locations (cf. Fig. 1c). (c) Evolution of the population density during 1300 weeks of simulation with scenario R3: periodic (every 6 months) hunting, random trap locations in the hunting localities used in 1995. (d) Evolution of the number of catches during 1300 weeks of simulation with scenario R1: continuous hunting and fixed trap locations (cf. Fig. 1c). (e) Evolution of the number of catches during 1300 weeks of simulation with scenario R2: periodic (every 6 months) hunting with the same trap locations (cf. Fig. 1c). (f) Evolution of the number of catches during 1300 weeks of simulation with scenario R3: periodic (every 6 months) hunting, random trap locations in the hunting localities used in 1995.

4.2. Hunting experiments

The five hunting experiments were run for 1300 weeks, i.e. 25 years. For all the experiments, the population density, total number of cells trapped and total number of catches were recorded each week. For each scenario, two charts show the evolution of population density and total number catches (cf. Figs. 5 and 6).

The charts for population density display the ten curves, whereas the chart showing total number of catches displays the average weekly values calculated from the corresponding ten runs. The first hunting season is unusual in that it always has a drastic effect on the population. This is not shown on the charts for number of catches because it cannot be accommodated on the Y-axis with the chosen scale.

Some indicators are summarized in Tables 3 and 4 for the first and second set of experiments, respectively. The first indicator is hunting coverage, expressed in number of cells, which is an indication of the global hunting pressure. The second is the population density after 25 years, which indicates the population's viability for each scenario. The last indicator is the total number of catches during the 25 years, which gives an idea of how efficient the different scenarios are at providing wild meat for the hunters.

For the first set of experiments, the global hunting pressure is the same for the three scenarios (Table 3). The results show that the popula-

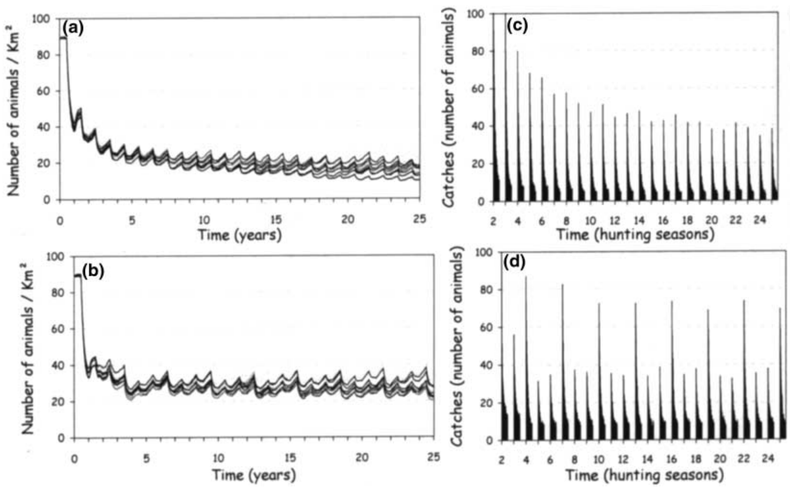

Fig. 6. (a) Evolution of the population density during 1300 weeks of simulation with scenario H1: 90 hunters acting individually (individual turnovers) with a set of four given trap networks. (b) Evolution of the population density during 1300 weeks of simulation with scenario H2: 30 groups of three hunters (collective turnovers) with a set of four given trap networks. (c) Evolution of the catches during 1300 weeks of simulation with scenario H1: 90 hunters (individual turnovers) with a set of four given trap networks. (d) Evolution of the number of catches during 1300 weeks of simulation with scenario H2: 30 groups of three hunters (collective turnovers) with a set of four given trap networks.

Table 3
Global results (mean values and S.D. in brackets) for the first set of experiments (repeated 1995 locations)

Experiments	Hunting coverage (number of cells)	Population density after 25 years (number of animals/km^2)	Total catches during 25 years (number of animals)
R1 — continuous hunting, fixed trap location	722.9 (4.7)	33.64 (1.8)	14 321.9 (405.7)
R2 — periodic hunting, fixed trap location	718.6 (7.4)	38.20 (1.1)	16 036.8 (346.5)
R3 — periodic hunting, random trap location in the hunting localities	717.3 (9.2)[a]	23.62 (0.7)	13 400.9 (219.6)

[a] Calculations based on $25 \times 10 = 250$ data, otherwise calculations are based on 10.

Table 4
Global results (mean values and S.D. in brackets) for the second set of experiments (with hunter agents)

Experiments	Hunting coverage (number of cells)	Population density after 25 years (number of animals/km^2)	Total catches during 25 years (number of animals)
H1 — Hunters' periodic individual turnovers	640.8 (28.4)[a]	15.77 (3.4)	12 073.7 (766.1)
H2 — Hunters' periodic collective turnovers	747.2 (26.5)[a]	24.21 (3.4)	13 970.3 (824.3)

[a] Calculations based on $25 \times 10 = 250$ data, otherwise calculations are based on 10.

tion density decreases rapidly over 6 years and then reaches an equilibrium state that is stable for experiment R1 but periodic for R2 and R3 (Fig. 5a–c). Before the population density stabilizes in scenarios R1 and R2, it drops to a minimum value that is below the equilibrium value. However, this is not the case for scenario R3. This is probably an artificial phenomenon, caused by the fact that traps are set in the same locations, which can also be seen on the charts for number of catches (Fig. 5d–f).

The impact of the hunting periodicity is clearly shown by comparing the population densities after 25 years — in R2, there are about 5 animals/km^2 more than in R1 — and by comparing the total number of catches during 25 years (Table 3).

It is interesting to compare this with the impact of relocating trap networks in the hunting localities (R3). Here, the population density in R2 is around 15 animals/km^2 more than in R3. This result suggests that the spatial location of traps is much more important than their periodicity. The impact on the number of catches is similar if we look at the global results (Table 3). When Fig. 5e and f are examined in detail, we can see that for both R2 and R3, about 50 animals were captured during the first few weeks of the hunting season, whereas fewer animals were caught in the last week, i.e. only about 15 animals in R2 and less than ten animals in R1.

In the second set of experiments, the H1 scenario is clearly the worst: after 25 years, the population (Fig. 6a) and number of catches (Fig. 6c) are still decreasing. The hunting coverage is low (Table 4), which suggests that there could be a problem of spatial congestion ($90 \times 4 = 360$ individual trap networks have to be identified, some of which overlap). The result is that some hunter agents are unable to access their traps because the space is already occupied by other agents. There may be another reason to explain why there is a difference of about 9 animals/km^2 between the population density of H1 and H2 (Table 4). The lack of coordination between the hunter agents for scenario H1 should erase the effect of individual turnovers. Fig. 6b and d illustrate the positive impact of the rules that govern collective turnovers (in the case of period-3 synchronized collective turnovers).

It is surprising to note that the overall results from these two sets of experiments (Tables 3 and 4) show that the best scenario seems to be R2 in which trap networks stay in the same location. Two observations can be made. Firstly, Fig. 1c shows that not all hunting localities were used in 1995. These areas act as a hunting preserve in scenario R2. In the H2 experiment, this effect does not occur because all 27 hunting localities are distributed to establish the $30 \times 4 = 120$ trap networks.

Furthermore, the global indicator for the number of catches is expressed in number of animals. It does not take into account the size of trapped animals. In addition, given that only young animals move around, there is likely to be a qualitative difference in the age distributions of animals caught when traps are in a fixed location (R2) and when trap location is varied (H2). Fig. 7b and c show the relative proportions of the age classes in the total population just before the 25th hunting season. Compared to Fig. 7a, these distributions illustrate the impact of hunting. However, it is important to note that they are similar: in experiment R2, 36.9% of animals caught were big (aged 4–7 years) compared to 37.3% in H2.

Fig. 8a and b show the relative proportion of the age classes in the caches of last (25th) hunting season for scenario R2 and H2, respectively. These results confirm the above supposition, namely that the proportion of big animals caught (age classes 4–7 years) in the H2 scenario is double that of R2, 26.2 and 13.6% respectively.

5. Discussion

This chapter is divided into three parts. Firstly, we draw conclusions about the way hunters in Djemiong manage the resource. Secondly, we discuss multiagent methodology and, lastly, we propose an approach to help the inhabitants understand the implications of the MAS results.

5.1. Resource management in Djemiong

In Africa, wild animals are a vital source of dietary protein for humans. Since the beginning of the 20th century, protected areas have been considered as an effective way of preserving re-

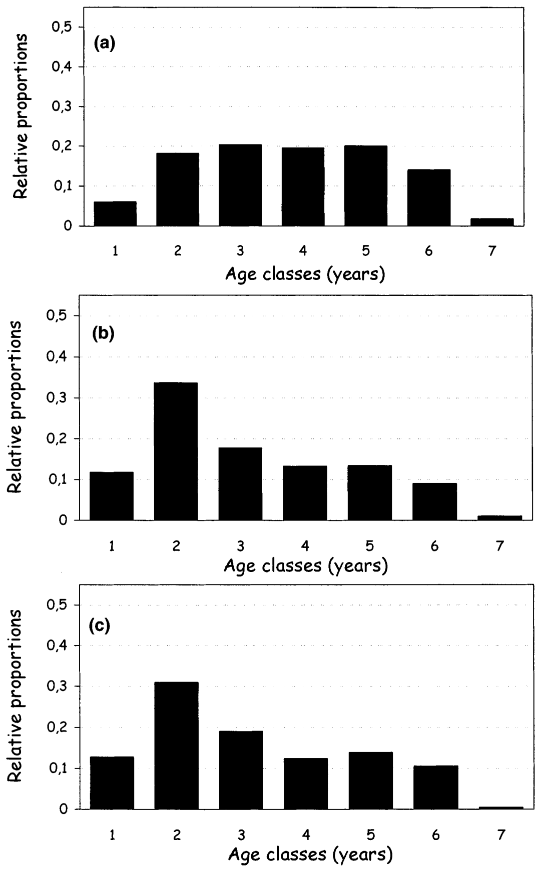

Fig. 7. (a) Age structure of the population after 5000 weeks of simulation with no hunting. (b) Age structure of the population after 1300 weeks of simulation with scenario R2: periodic (every 6 months) hunting with the same trap locations (cf. Fig. 1c). (c) Age structure of the population after 1300 weeks of simulation with scenario H2: 30 groups of three hunters (collective turnovers) with a set of four given trap networks.

sources. The major problem associated with protected areas is the fact that they exclude local populations who traditionally use the resource. Protected areas are important for food and the economy but they also have an important sociocultural role. They are the cause of a number of conflicts in different parts of the world. Authorities impose certain constraints and apply sanctions in order to prevent the depletion of resources in protected areas. Traditional activities carried out by local people are often made illegal. Many people argue that local populations should be consulted before a protected area is designated. In this way, it would be possible to determine whether or not they were already managing their resources. This should include a description of local management schemes and a discussion of their function. Modelling is a useful method of testing how management rules operate.

In the case of Djemiong, the experiments described here demonstrate the importance of the spatial dimension. The classical models of exploited population dynamics (for example in fishery science) often fail to take this aspect into account. The hypothesis concerning the local management strategy for the blue duiker was established before the simulations were carried out. It was based on observations and discussions with the hunters (seasonality of the hunting activity). This seasonality is similar to the management

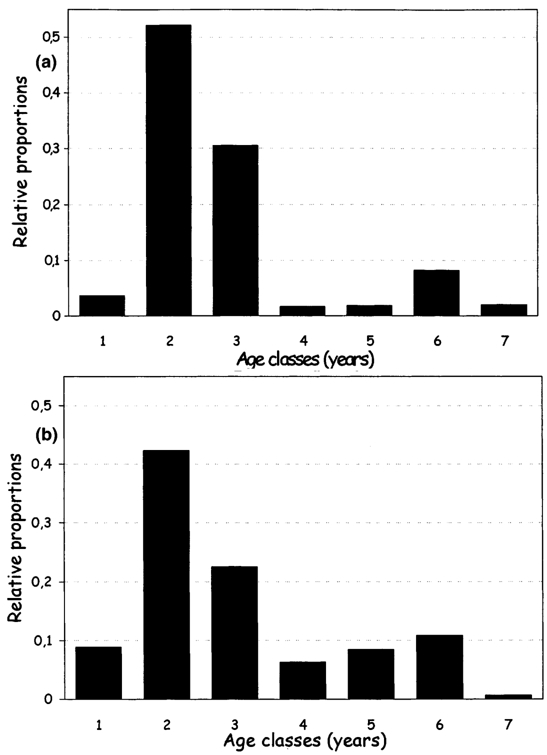

Fig. 8. (a) Age of the animals caught during the 25th hunting season of simulation with scenario R2: periodic (every 6 months) hunting with the same trap locations (cf. Fig. 1c). (b) Age of the animals caught during the 25th hunting season of the simulation with scenario H2: 30 groups of three hunters (collective turnovers) with a set of four given trap networks.

rules adopted in other countries to protect renewable resources. From the simulation, it seems that the dynamics of the spatial distribution of the catching effort are due to causality. By studying this phenomenon, we observed that the rotation is effective if there is some coordination between the agents. This coordination is the basis of society's social structure. In Djemiong, the position of the traps is determined by social factors. Social subgroups have preferential access to parts of the forest and each year they decide collectively where to hunt. These places are part of the family heritage. Thus, resource management is more complex than it seems, i.e. it has to be understood in the context of the links between social organization and spatial structure. In order to deal with this degree of complexity, multiagent simulation is used to identify the relevant level of organization (groups rather than individuals) and the appropriate spatial scale (network rather than hunting localities).

5.2. Multiagent simulations

In the introduction, we claim that MAS are effective tools for implementing and simulating individual-based models and that they also provide a more general framework for studying the organization of agents. This can be discussed in the context of the experiments described here. Two versions of the hunting model are proposed: one with hunter agents that perceive and act on their environment, and one involving groups of agents that can coordinate the actions of several hunter agents. This aspect of the model, in particular, demonstrates the value of multiagent systems. A MAS can deal with the problems associated with agents' interactions on different scales (not just that of the individual) and can model both the autonomy of the agents and their organization. We assume that the dynamics of the system have to be understood in terms of the balance between autonomy and organization. The same assumption applies to only a few IBMs in ecology that deal with interactions across trophic levels. Very few simulate the interactions between agents on different levels, e.g. the individuals and the group.

In his paper on the complexity of ecosystems, Holling (1987) proposes three concepts that dominate perceptions of ecological causation, behaviour and management. The first concept is constant nature and the second is resilient nature. The latter emphasizes variability and non-linear causation due to small-scale events, which is the case for most IBMs. The third concept is one of organizational change (nature evolving). This relates to concepts of function and organization that influence how elements are connected. Recent research on hierarchy theory (Allen and Hokstra, 1992) proposes an approach to ecological questions that involves the study of the interactions between different levels: organism, populations, communities, ecosystems, landscapes and biomes. "Apart from organism and biosphere levels, there is plenty of room for entities from almost any type of ecological system to be contained within an entity belonging to any other class of the system." We consider that MAS is an appropriate method for modelling the organization of entities that interact on different scales.

5.3. Models and the participative approach

The trend now is to encourage conservation/participation as opposed to preservation. Participatory projects attempt to link sustainable resource management and local development (integrated conservation-development projects). The participative approach does not indicate the 'right way' to preserve resources, but rather encourages stakeholders to cooperate so that they are all involved in the decision-making process.

It can be difficult for a scientists with their own cultural prejudices to fathom how resources are managed in a situation where resource exploitation is determined by local and customary practices. Similarly, it can be hard for a manager to explain the reasons for his management choices to local people and to persuade them to accept the decisions. The participative approach is based on the principle that each party that uses a natural resource should be able to negotiate its own future. Communication is extremely important for this. The first step involves finding a practical common language for expressing the different viewpoints. The next step is to compare the potential consequences of the different viewpoints, for which an appropriate tool is needed. The chosen tool should be able to demonstrate the usefulness of a given proposal. Generally, mathematical equations are not suitable for this purpose. Our aim is to test the pertinence of MAS in the context of a participative approach. MAS have the potential to build a shared representation of a system that can then be used to test different scenarios drawn up collectively.

We need to define a method to ensure that all parties perceive the chosen model as an acceptable common representation of the system. A computer screen displaying small dots that move around a spatial grid made up of coloured cells can be as alien as a mathematical equation. Therefore, as an intermediate step, we recommend subjecting MAS to a social validation test, i.e. to see whether the people concerned recognize the relevance of the agents that represent them in the model. Instead of computer demonstrations, we propose organizing a role play session that allows the people to play a given role (defined as the translation of the corresponding agent in the MAS). This methodology is currently being tested in the context of a study of irrigated schemes in Senegal. The preliminary results are promising.

Once the MAS has been accepted as a legitimate tool, the idea is to use it as a guide during the negotiation process. The MAS should be accepted when a comparison has been made between a run with the model and the situation at the end of the role play.

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